KEVIN LIANG

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EDUCATION

Master of Applied Science in Electrical and Computer Engineering **Sep 2025 – Aug 2027**
University of Toronto *Toronto, Ontario, Canada*

- **Supervisor:** Prof. Lacra Pavel
- **Scholarships & Awards:** Edward S. Rogers Sr. Graduate Scholarship

Bachelor of Applied Science in Mathematics and Engineering with Professional Internship **Sep 2020 – Apr 2025**
Queen's University *Kingston, Ontario, Canada*

- **Supervisor:** Prof. Fady Alajaji
- **Scholarships & Awards:** First Class Honours, Science '66 Memorial Prize, Mildred K. Walters Awards

RESEARCH EXPERIENCE

Undergraduate Thesis **Sep 2024 – Apr 2025**
Queen's University *Kingston, Ontario, Canada*

- **Thesis Title:** Speech Decomposition Using the Variational Information Bottleneck Method
- **Supervisor:** Prof. Fady Alajaji
- Designed and applied the Variational Information Bottleneck (VIB) method to the SpeechSplit model, improving decomposition performance for timbre and pitch components

Summer Research Student **May 2022 – Aug 2022**
The Materials at Extremes Nuclear Research Institute *Kingston, Ontario, Canada*

- **Supervisor:** Prof. Mark Daymond
- Optimized and programmed features of a beam energy activity calculator with Java and SQL, reducing output error from more than 50% to less than 20% through a new algorithm
- Developed SRIM/TRIM automation software and an energy optimizer with Python to conduct uniform helium beam research, reducing calculation error to 5% and shortening the time required for ion irradiation experiments with a particle accelerator
- Generated 3 technical reports and relevant software documentation for inclusion in publications

TEACHING EXPERIENCE

Teaching Assistant **Sep 2026 – Dec 2026**
University of Toronto *Toronto, Ontario, Canada*

- ECE311H1 (Introduction to Control Systems): tutorials and marking

WORK EXPERIENCE

Product Management Analyst **Sep 2023 – Aug 2024**
RBC Capital Markets – Electronic Trading Execution Management *Toronto, Ontario, Canada*

- Implemented, tested, and deployed 30+ custom trading algorithms per month for clients in NA and EMEA equities markets
- Pitched, developed, and productionized an internal analytics web platform using Flask to process and analyze client and algorithm data for insights. Used by 20+ staff members across management, product management, and client coverage teams
- Wrote and deployed 4 Python scripts to produce analytic reports, saving team members 2+ hours daily

Robotics Engineer – Special Projects **May 2021 – Aug 2021**
Wizrobotics *Richmond Hill, Ontario, Canada*

- Managed a team of 3 to create a new summer school program that taught Roblox Lua and attracted 20+ customers
- Developed 4 teaching curriculums for Thunkable, Code.org, Roblox Lua, and App Inventor using Twine

SCHOLARSHIPS & AWARDS

Edward S. Rogers Sr. Graduate Scholarship, University of Toronto **Apr 2025**

- Awarded as part of the Master of Applied Science program at the University of Toronto

Canadian Hyperloop Conference 2023 Leadership Award, Hyperloop Global **May 2023**

- Awarded for leading Queen's Hyperloop Design Team, the top-ranked team at the Canadian Hyperloop Conference 2023, throughout the 2022 – 2023 season

Science '66 Memorial Prize, Engineering Society at Queen's University **Mar 2023**
- Awarded for contributing the most to engineering extracurricular activities during my first 3 years at Queen's

Mildred K. Walters Awards, Queen's University **Dec 2022**
- Awarded for financial need and academic achievement

11 Competition Awards, Various Local, Provincial, and National Competitions **Nov 2021 – Nov 2024**

LEADERSHIP EXPERIENCE

Co-Captain, Logistics Lead **Apr 2023 – Mar 2025**
Queen's Aerospace Design Team *Kingston, Ontario, Canada*
- Co-led 120+ members across 3 branches to build autonomous drones and produce research papers to compete in 3 international competitions. Raised over CAD 30,000 and won 1 award for best internal processes

Senior Project Manager, VP of Technology, Project Manager **Mar 2022 – Mar 2025**
Queen's Startup Consulting *Kingston, Ontario, Canada*
- Managed and trained a team of 6 consultants to provide ML-based sales prediction and marketing services to a women's boxing brand founded by a NEXT Founder

PD Workshops Chair, Discipline Club VP of External, EngPal Mentor, First Year Representative **Oct 2020 – Mar 2025**
Engineering Society at Queen's University *Kingston, Ontario, Canada*
- Mentored 5 first-year engineering students and partnered with the faculty's corporate relations team to host 10+ professional development workshops and events

Advisor, Co-Captain, Chassis Manager, First Year Representative **Sep 2020 – Nov 2024**
Queen's Hyperloop Design Team *Kingston, Ontario, Canada*
- Co-led 160+ members and managed 7 projects to build Hyperloop pods, produce research papers, and design tunnel boring machines to compete in 2 international competitions. Raised CAD 80,000 and won 8 awards at competitions

Director of UI/UX, UI/UX Designer, Software Developer **Mar 2021 – Mar 2024**
Queen's Technology and Media Association *Kingston, Ontario, Canada*
- Collaborated with 2 developer teams to develop, market, and launch software products. Pitched and won 1st place at the annual product competition co-hosted by McKinsey & Company

TECHNICAL SKILLS

- **Programming:** Python, MATLAB, C++, C, Java, HTML, CSS, JavaScript, SQL, kdb+/q, VHDL, Assembly
- **Tools:** Microsoft Excel, Microsoft Word, Microsoft PowerPoint, Jira, Jenkins, Git, Flask, Selenium, OpenCV, TensorFlow, MongoDB, Express.js, React.js, React Native, Node.js, Bootstrap, Expo, Maple, LaTeX
- **Design:** Figma, Adobe Creative Cloud, SOLIDWORKS CAD

ADDITIONAL INFORMATION

- **Languages:** English (Native), Mandarin (Native)
- **Interests:** Books, painting, chess, board games, and finding the best truffle fries in the world (it's in Toronto so far)

REFERENCES

Lacra Pavel, University of Toronto **pavel@control.utoronto.ca**
- Professor, Electrical and Computer Engineering

Fady Alajaji, Queen's University **fa@queensu.ca**
- Professor, Mathematics and Engineering

Weijian Li, University of Notre Dame **wli26@nd.edu**
- Postdoctoral Researcher, Electrical and Computer Engineering